William Castro

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EDUCATION

Boston University, College of Engineering — Boston, MA
Master of Science in Mechanical Engineering — *Candidate, December 2026*
Late Entry Accelerated Program (LEAP) — *September 2024 - Present*

University of Massachusetts Amherst, Isenberg School of Management — Amherst, MA
Bachelor of Business Administration in Operations and Information Management — *Graduated, May 2023*

- GPA 3.86; Dean's List All Semesters, Cum Laude Honors, Data Analytics Certificate

RESEARCH EXPERIENCE

Lubner Group, Boston University — Boston, MA
Direct Air Capture of CO_2 – Laboratory Assistant — *May 2026 - Present*

- Building hardware for the lab's planned 5,000-cycle degradation test of metal-organic frameworks (MOFs), porous materials that capture CO_2, where heating and cooling time per cycle sets how long the test takes
- Designed and CNC milled an aluminum vacuum chamber, with a waterjet-cut lid and laser-cut nitrile gasket
- Reached 0.8 mbar and traced leaks to the NPT fittings, not the chamber or gasket, using isopropanol testing; now switching to Swagelok fittings to complete vacuum validation
- Designed and 3D printed an air funnel for a 120 mm fan, cutting tube cooldown from 160 °C to 35 °C from ~10 min to ~4 min (2.5x faster)
- Built and tested a second tube furnace core with thicker nichrome wire that reached 90 °C in under a minute, about twice as fast as the original
- Machined aluminum sample holders to ~0.0015" clearance in the tube bore, preventing powder blowout

PROJECTS

Rotary Vane Pump — *September 2026 - Present*

- Building a rotary vane pump in BU's Manufacturing Processes course using sand casting, CNC and manual machining, and drilling and tapping, working to the drawing specifications
- Bored the rotor hole to 0.7480" +0.0005/−0.0000, verifying it with a custom stepped plug gauge since no available tool could measure that tolerance, so the shaft could be shrink fit into it; the fit was successful
- Redesigning the pump with a team of 3 for liquid ammonia (−50 °F, 8 atm, 3,600 RPM, 4,500-hour MTBF), producing GD&T drawings detailed enough for manufacture

PressMate Coffee Press — *January 2026 - May 2026*

- Designed the PressMate, a steel coffee press, with a 4-person team in BU's Product Realization course, balancing a limited budget, safety, and customer use-case requirements
- Built 4 mockups and 14 steel prototypes (1 EVT, 3 DVT, 10 PVT) using CNC and manual lathes, waterjet cutter, TIG welding, cast silicone, and wire EDM in a 22-step process for each prototype
- Optimized the CNC lathe process for steel tube by testing inserts, boring bars, cut depths, speeds, feeds, and chuck clamping force, significantly improving surface finish
- Diagnosed plunger binding in the PVT as an O-ring groove that was too narrow, not surface finish, and fixed it with chamfers; every unit sealed and fit correctly
- Created and maintained 16 controlled documents, including drawings, process plans, SOPs, tooling, RASCI, risk management, and cost model, with templates and revision tracking through EVT, DVT, and PVT

Automated Circuit Printing — *January 2026 - May 2026*

- Expanded a robotic cell of two CNC mills and three UR5 arms with a 6-person team, adding a potentiometer and buzzer to produce 2 functional circuits from milled HDPE and silver conductive ink
- Extended the Python control code, automated CAD/CAM, and pick-and-place programs for the new components; designed and milled an HDPE component tray
- Tuned UR5 pick-and-place in sub-millimeter increments so components landed accurately on the board
- Fixed uneven conductive-ink extrusion with a larger syringe tip; added a 2–3 mm code offset to protect the tool

Thermal Expansion Simulation of a Pump Housing *September 2025 - December 2025*

- Modeled a gray iron rotary vane pump housing in COMSOL (heat transfer coupled with solid mechanics) with a team of 3, building the geometry, constraints, and boundary conditions
- Predicted ~0.0015" bore growth from 70 °F to 140 °F and validated it against a hand calculation, showing thermal expansion was not significant for the pump design

Accumulator Cart for High Voltage Battery, Terrier Motorsports *November 2024 - February 2025*

- Led a 4-person team in designing a hydraulic accumulator cart to safely remove a high-voltage battery from a racecar, prioritizing ease of use and adaptability to future battery designs
- Incorporated a deadman brake, electrical insulation, and an integrated battery charger into the design to protect operators and improve usability
- Designed new wheel brackets to fit the brakes, validating a 500 lb load capacity with SolidWorks FEA

WORK EXPERIENCE

Ernst & Young (EY) Boston, MA

Client Serving Intern, Technology Risk Consulting *June 2022 - August 2022*

- Completed SOX control testing for various clients, including change and access controls, to ensure compliance
- Documented audit walkthroughs and testing results to support audit readiness and adherence to standards
- Supported engagement teams with project planning and client deliverables

Launch Intern *June 2021 - August 2021*

- Tested controls to ensure design and operating effectiveness
- Rotated between consulting, audit, and tax client-serving teams
- Worked with fellow interns on a capstone project to help solve a Tennessee labor problem

Miranda Bread Marlborough, MA

Operations and Maintenance Coordinator *Summer 2017 - January 2023*

- Maintained all company vehicles, including two Sprinter vans
- Handled day-to-day repairs and replacements of broken equipment to keep each bakery running
- Troubleshot network, POS system, printer, and office computer issues across locations
- Managed inventory deliveries to store locations and tracked inventory across locations and a production plant
- Reconciled receipts for money transfer services worth over $100,000 in a single shift
- Monitored store cleanliness and employee conduct across locations

SKILLS

Engineering Software: SolidWorks, Fusion 360, Onshape, COMSOL, FEA, Python, MATLAB
Data & Business Tools: SQL, R, JMP, Tableau, Power BI, VBA, SAP, Microsoft Excel
Manufacturing: CNC Mill, CNC Lathe, Manual Mill, Manual Lathe, Waterjet Cutter, Laser Cutter, 3D Printing, TIG Welding, Sand Casting, Soldering
Language: Conversational in Portuguese

LEADERSHIP & ACTIVITIES

Operations & Information Management Club, Association of Latino Professionals for America

Treasurer, Fundraising Chair *September 2019 - May 2023*

- Managed club finances, including the bank and Venmo accounts, and created yearly budgets
- Organized fundraisers to make sure the club had the funds to operate
- Connected with recruiters, professionals, and alumni in the operations and information management field